7.1: Design Requirements and Constraints

Item	Entry	Source/Justification
Parameters to be Identified	Motor Resistance (R_m), Motor back-emf constant (k_m)	What the experiment is determining
Quantities to measure, and block/instrument used for each	Motor Current ($i_m(t)$); analog measurement Angular Speed ($\omega_m(t)$); using encoder for position, then compute for speed Input Voltage ($v_m(t)$); through input set using constants	Used to calculate R_m and k_m . The input voltage can be set, and the current and speed can be measured using blocks in Simulink, which can be used within formulas
Maximum voltage to apply	5 V	Applying more than 5 V could lead to damage of the motor, especially if it is stalled for a period of time
Stall constraint: maximum voltage and maximum duration for which shaft may be held stationary	4 V 10 s	If motor is stalled at high voltages (>5 V) for long periods of time (>10 s), then it can lead to the motor being damaged
Other constraints to protect hardware	Keep all loose items away from the motor. Stall detection block. Current Limit: 0.476A for Experiment 1	Detects if the voltage is greater than 5 V and if the motor is stalled for longer than 10 s and stops the code.
Equipment limits that bounds measurement	Position encoder resolution 2048/rev 12-bit ADC for current	These are mechanical limits that restrict the accuracy of the speed and current measured
# of trials and repetitions per trial	Experiment 1: 1 trial, 3 repetitions Experiment 1: 2 trials, 3 repetitions	Multiple trials and repetitions should be done, but with the constraint of finishing the lab in 30 minutes, it should not be excessive
Assumptions	Assume the inductance of the motor, L_m, can be ignored.	The inductance of the motor is much less than its resistance and can be ignored in the KVL equation.

7.2: Experiment Design

Experiment #1:

Summary:

We will be leveraging the KVL equation, $v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0$ to determine the value of R_m. The experiment involves stalling the motor to retrieve values of i_m to find out R_m:

IMPORTANT: Max Input Voltage is 10V. Do not run the motor over 5V for stalling. Do not stall the motor for more than 10s.

PROCEDURE

Initialization:

1. Plug in the QUBE-Servo 2 into the computer via USB and connect it to power using the dedicated power cable.
2. Load the MATLAB software. Create a new SIMULINK file.
3. Insert the HIL Initialize block.
4. Double click on the HIL Initialize block, and in the Board type field, select qube_servo2_usb.
5. Under the QUARC tab, select quarc_win64.tlc as the QUARC Target. After this, you should have a new tab called "HARDWARE".
6. In the Hardware tab, click Monitor & Tune. If your system has been successfully connected, the status LED on the Qube Servo will turn green.

Determining $i_m(t)$:

1. Use a constant block for the input voltage, $v_m(t)$, to write a value between 1V-4V into the HIL Write block. Record the input voltage value, $v_m(t)$ that you used in the table below.
2. Add a HIL Read block and set it to a0 and connect it to a Display block. This will measure and record the current, $i_m(t)$, through the circuit in Amps.
3. Add a second HIL Read block and set it to e0. This is for the rotational position, which will be used for the "Stall Detection Block"
4. Insert the "Stall Detection Block", as previously done in 6.3.5 of the lab manual. The inputs to this block are the input voltage signal and the rotational position output. This is important to include so that the motor does not stall out for too long.
5. Set the time duration to 10 seconds.
6. Run the program. After around 1 second, stall the motor by gripping the sides of the red disk. Hold it in place until the program terminates.
7. After the program finishes, record the final current value $i_m(t)$ read into the table below.
8. Repeat steps 1-7 with the same input voltage and record the voltage and current values below.

Repetitions	Voltage Input (V)		Current Output (A)	
1	$v_m(t)$		$i_m(t)$	
2			$i_m(t)$	
3			$i_m(t)$	
Average: $x_{average} = \frac{x_1 + .. x_n}{n}$	$v_m(t)$		$i_m(t)$	

Determining R_m:

Using the KVL equation defined in the summary, isolate an expression for R_m and use the average values from the above table to solve for R_m.

$$R_m = \frac{v_m(t)}{i_m(t)} \qquad (1)$$

$$R_m = _____$$

Experiment #2:

Summary of Experiment:

We will be leveraging the KVL equation, $v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0$ to determine the value of k_m. The experiment involves using two different input voltages and a system of equations to find out k_m:

$$v_{m(1)}(t) - R_m i_{m(1)}(t) - k_m \omega_{m(1)}(t) = 0$$
$$v_{m(2)}(t) - R_m i_{m(2)}(t) - k_m \omega_{m(2)}(t) = 0$$

PROCEDURE

IMPORTANT: Max Input Voltage is 10V. Do not run the motor over 5V for stalling. Do not stall the motor for more than 10s.

Initialization:

1. Plug in the QUBE-Servo 2 into the computer via USB and connect it to power using the dedicated power cable.
2. Load the MATLAB software. Create a new SIMULINK file.
3. Insert the HIL Initialize block.
4. Double click on the HIL Initialize block, and in the Board type field, select qube_servo2_usb.
5. Under the QUARC tab, select quarc_win64.tlc as the QUARC Target. After this, you should have a new tab called "HARDWARE".
6. In the Hardware tab, click Monitor & Tune. If your system has been successfully connected, the status LED on the Qube Servo will turn green.

Determining $\omega_m(t)$ and $i_m(t)$:

1. Use a constant block for the input voltage, $v_m(t)$, to a value between 1V-4V into the HIL Write block. Record the input voltage value, $v_{m(1)}(t)$ that you used in the table below.
2. Add the HIL Read Encoder block (make sure you set it to e0 to read the number of counts measured by the encoder) and convert the angular position to radians using Simulink Blocks, as done previously in "6.2: Read the Encoder".
3. Add a second HIL Read block and set it to a0 (reads the current measurement in A) and connect it to a **Display block**.
4. Feed the Angular Position (output from Step 2) to a **Derivative block** to give the Angular Speed (rad/s) as done previously in "6.4: Filtering". Connect the output of the derivative block to a **Display Block.**
5. Under the "HARDWARE" tab, click "Monitor & Tune" and record $i_{m(1,1)}(t)$ and $\omega_{m(1,1)}(t)$ for your first iteration. You will then click "Monitor & Tune" with the same input voltage and record $i_{m(1,2)}(t)$ and $\omega_{m(1,2)}(t)$. Repeat this step for a third time, click "Monitor & Tune" and record the values for $i_{m(1,3)}(t)$ and $\omega_{m(1,3)}(t)$. Determine the average current value ($i_{m(1)}(t)$) and average speed ($\omega_{m(1)}(t)$) from your recorded measurements and record these in Table 1 as well.
6. Repeat Steps 1-5 under "Determining $\omega_m(t)$ and $i_m(t)$" for a different input voltage value, $v_{m(2)}(t)$, average current, $i_{m(2)}(t)$, and average speed, $\omega_{m(2)}(t)$. **Record these values in Table 2 below.** **SEE NEXT PAGE FOR STEP 7!**

7. You will now have values for $v_{m(1)}(t)$, $i_{m(1)}(t)$, $\omega_{m(1)}(t)$, $v_{m(2)}(t)$, $i_{m(2)}(t)$ and $\omega_{m(1)}(t)$. With these 6 values, we can determine the value for for k_m using the following formula (Eqn, 2):

$$k_m = \frac{\left[\frac{v_{m(2)}(t)}{i_{m(2)}(t)} - \frac{v_{m(1)}(t)}{i_{m(1)}(t)}\right]}{\left[\frac{\omega_{m(2)}(t)}{i_{m(2)}(t)} - \frac{\omega_{m(1)}(t)}{i_{m(1)}(t)}\right]} \quad (2)$$

Trial 1: $v_{m(1)}(t)$ = _____

Repetition #	Current Output (A)		Angular Speed Outputs (rad/s)	
1	$i_{m(1,1)}(t)$		$\omega_{m(1,1)}(t)$	
2	$i_{m(1,2)}(t)$		$\omega_{m(1,2)}(t)$	
3	$i_{m(1,3)}(t)$		$\omega_{m(1,3)}(t)$	
Averages: $x_{average} = \frac{x_1 + \ldots x_n}{n}$	$i_{m(1)}(t)$		$\omega_{m(1)}(t)$	

Trial 1: $v_{m(2)}(t)$ = _____

Repetition #	Current Output (A)		Angular Speed Outputs (rad/s)	
1	$i_{m(2,1)}(t)$		$\omega_{m(2,1)}(t)$	
2	$i_{m(2,2)}(t)$		$\omega_{m(2,2)}(t)$	
3	$i_{m(2,3)}(t)$		$\omega_{m(2,3)}(t)$	
Averages: $x_{average} = \frac{x_1 + \ldots x_n}{n}$	$i_{m(2)}(t)$		$\omega_{m(2)}(t)$	

k_m = _____

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Derivation of Equation to Solve Experiment #1:

From the experiment, by stalling the motor, the speed $\omega_m(t)$ is 0 rad/s, so the term $k_m\omega_m(t)$ is equal to 0. From there, we can isolate for R_m and solve it with the recorded values:

$$v_m(t) - R_m i_m(t) - k_m\omega_m(t) = 0,\ \omega_m(t) = 0$$

$$v_m(t) - R_m i_m(t) = 0$$

$$v_m(t) = R_m i_m(t)$$

$$\boxed{R_m = \frac{v_m(t)}{i_m(t)}}$$

Derivation of Equation to Solve Experiment #2:

How to solve it:

From the first equation:

$$v_{m(1)}(t) - k_m\omega_{m(1)}(t) = R_m i_{m(1)}(t)$$

$$\frac{1}{i_{m(1)}(t)}[v_{m(1)}(t) - k_m\omega_{m(1)}(t)] = R_m$$

Using a similar relationship derived from the second equation,

$$\frac{1}{i_{m(2)}(t)}[v_{m(2)}(t) - k_m\omega_{m(2)}(t)] = R_m$$

$$\frac{1}{i_{m(2)}(t)}[v_{m(2)}(t) - k_m\omega_{m(2)}(t)] = \frac{1}{i_{m(1)}(t)}[v_{m(1)}(t) - k_m\omega_{m(1)}(t)]$$

$$\frac{1}{i_{m(2)}(t)}v_{m(2)}(t) - \frac{1}{i_{m(2)}(t)}k_m\omega_{m(2)}(t) = \frac{1}{i_{m(1)}(t)}v_{m(1)}(t) - \frac{1}{i_{m(1)}(t)}k_m\omega_{m(1)}(t)$$

$$\left[\frac{1}{i_{m(2)}(t)}v_{m(2)}(t) - \frac{1}{i_{m(1)}(t)}v_{m(1)}(t)\right] = k_m\left[\frac{1}{i_{m(2)}(t)}\omega_{m(2)}(t) - \frac{1}{i_{m(1)}(t)}\omega_{m(1)}(t)\right]$$

$$k_m = \frac{\left[\frac{1}{i_{m(2)}(t)}v_{m(2)}(t) - \frac{1}{i_{m(1)}(t)}v_{m(1)}(t)\right]}{\left[\frac{1}{i_{m(2)}(t)}\omega_{m(2)}(t) - \frac{1}{i_{m(1)}(t)}\omega_{m(1)}(t)\right]}$$

$$\boxed{k_m = \frac{\left[\frac{v_{m(2)}(t)}{i_{m(2)}(t)} - \frac{v_{m(1)}(t)}{i_{m(1)}(t)}\right]}{\left[\frac{\omega_{m(2)}(t)}{i_{m(2)}(t)} - \frac{\omega_{m(1)}(t)}{i_{m(1)}(t)}\right]}}$$